Real-World Order of Operations

Question Bank - Additional 4.07

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

For questions 1 through 4, substitute and evaluate.

1. $5x^2$ when $x = -4$

2. $\frac{1}{2}mv^2$ when $m = 30$ and $v = 10$

3. $\frac{a^2+b}{c}$ when $a = 8$, $b = 4$, $c = 17$

4. $3(x+y)^2$ when $x = -5$ and $y = 2$

For questions 5 and 6, write and evaluate an expression. Give the unit.

5. A square room of side 13 feet is floored at 2.75 dollars a square foot, with a flat 90 dollars for delivery. Find the total cost.

6. A rectangular patio 16 feet by 7 feet is sealed at 1.40 dollars a square foot, with a flat 35 dollars for equipment hire. Find the total cost.

Two formulas.

7. Body mass index is $B = \frac{703w}{h^2}$, where w is a weight in pounds and h is a height in inches. Find B to the nearest tenth.

a. $w = 120$, $h = 62$

b. $w = 165$, $h = 68$

c. $w = 210$, $h = 74$

8. Kinetic energy is $K=\frac{1}{2}mv^2$, for mass m in kilograms and speed v in meters per second. Find K in joules.

a. $m=1{,}500$, $v=20$

b. $m=640$, $v=9$

c. $m=55$, $v=4$

Working backwards.

9. Find the missing value in each situation.

a. $\frac{1}{2}mv^2=1{,}600$ and $v=8$. Find m.

b. $x^2-7=74$, where x is positive. Find x.

c. A square lawn of side n feet is turfed at 3.00 dollars a square foot with a flat 55 dollars delivery, and the total is 1,255 dollars. Find n.

Reading the situation.

10. Two problems use the numbers 8, 12 and 4. **A:** a printer costs 8 dollars to hire plus 12 dollars for each of 4 days. **B:** a printer costs 8 dollars for each of 12 reams, plus 4 dollars for delivery. Find both totals, and explain why circling the numbers would not have told you what to do.

11. A square court of side 20 feet is resurfaced at 3.60 dollars a square foot, with a flat 200 dollars for the equipment. Which expression gives the total cost?

- ○ A. $20^2(3.60+200)$
- ○ B. $20(3.60)+200$
- ○ C. $20^2(3.60)+200$
- ○ D. $20^2+3.60+200$

12. Select ALL of the expressions whose value is 32 when $x=4$.

- ☐ A. x^2+16
- ☐ B. $\frac{128}{x}$
- ☐ C. $2x\cdot 2$
- ☐ D. $x^2\cdot 2$
- ☐ E. $(2x)^2$
- ☐ F. $2x^2$